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# 1. DINOv1 Theory (1/3): Self-Distillation with No Labels

DINO (**DI**stillation with **NO** labels) frames self-supervised learning as knowledge distillation from a momentum Teacher to a Student:

## Student–Teacher Architecture

- **Shared Network** $g = h \circ f$: Student ($g_{\theta_s}$) and Teacher ($g_{\theta_t}$) share a ViT/ResNet backbone $f$ and 3-layer MLP head $h$.
- **Projection Bottleneck**: Uses $\ell_2$-normalization and a weight-normalized linear layer ($K = 65536$ dims).

## Momentum Teacher Update (EMA)

- **Stop-Gradient on Teacher**: Gradients update only Student $\theta_s$; Teacher updates via EMA: $\theta_t \leftarrow \lambda\theta_t + (1-\lambda)\theta_s$.
- **Polyak-Ruppert Ensemble**: With $\lambda \in [0.996, 1]$, Teacher acts as a dynamic ensemble that guides the Student.

## Core Principle

The Student distills knowledge from an **evolving online Teacher** built from its own past iterations.

# 1. DINOv1 Theory (2/3): Multi-Crop & Distillation Objective

To encourage **local-to-global correspondence**, DINO extracts a set $V$ of multiple distorted views from each input image $x$:

## Temperature-Scaled Distillation Loss

- **Softmax Probability**:
$$P_s(x)^{(i)} = \frac{\exp(g_{\theta_s}(x)^{(i)}/\tau_s)}{\sum_{k=1}^{K} \exp(g_{\theta_s}(x)^{(k)}/\tau_s)}.$$
- **Cross-Entropy Loss**: Minimizes $\sum_{x \in \{x_1^g, x_2^g\}} \sum_{x' \in V \setminus \{x\}} H\big(P_t(x), P_s(x')\big)$ where $H(a, b) = -a \log b$.

## Multi-Crop View Assignment

- **Global Views** $\{x_1^g, x_2^g\}$: Two high-resolution crops ($224^2$, covering $> 50\%$ image area) are passed through **both Teacher and Student**.
- **Local Views**: Multiple small-resolution crops ($96^2$, covering $< 50\%$ area) are passed **only through the Student**.

## Local-to-Global Correspondence

Predicting Teacher's global context from Student's local patches forces the backbone to learn **high-level semantic context**.

# 1. DINOv1 Theory (3/3): Collapse Prevention & ViT Properties

DINO avoids representation collapse without contrastive negative pairs or predictor networks, unlocking unique properties in Vision Transformers:

## Collapse Prevention: Centering vs. Sharpening

- **Centering ($g_t(x) \leftarrow g_t(x) - c$)**: Prevents a single dimension from dominating by subtracting batch EMA center $c \leftarrow mc + (1-m)\frac{1}{B}\sum_{i=1}^{B} g_{\theta_t}(x_i)$.
- **Sharpening ($\tau_t \ll \tau_s$)**: Uses low Teacher temperature ($\tau_t = 0.04$, $\tau_s = 0.1$) to prevent collapse to a uniform distribution.

## Emergent Properties in DINO + ViT

- **Unsupervised Object Segmentation**: Self-attention maps of the `[CLS]` token automatically segment foreground object boundaries without labels.
- **Clustering for $k$-NN**: Frozen ViT features form well-separated semantic clusters, achieving strong $k$-NN classification ($78.3\%$ Top-1 on ImageNet).

## Key Takeaway from DINOv1

The synergy of **Self-Distillation + Momentum Teacher + Multi-Crop** makes ViT features naturally semantic and boundary-aware.

# 2. Motivation: Bottlenecks in Vanilla ViT (DINOv1)

In the standard DINOv1 baseline (`ViT-Small/16`), two core components introduce major bottlenecks:

## Bottleneck 1: Absolute Position Embeddings

- **Multi-Crop Distortion**: DINO jointly trains on $224^2$ (196 patches) and $96^2$ (36 patches). Bicubic interpolation alters distance metrics and distorts patch aspect ratios.
- **No Translation Invariance**: Fails to directly model continuous relative displacements $(\Delta x, \Delta y)$.

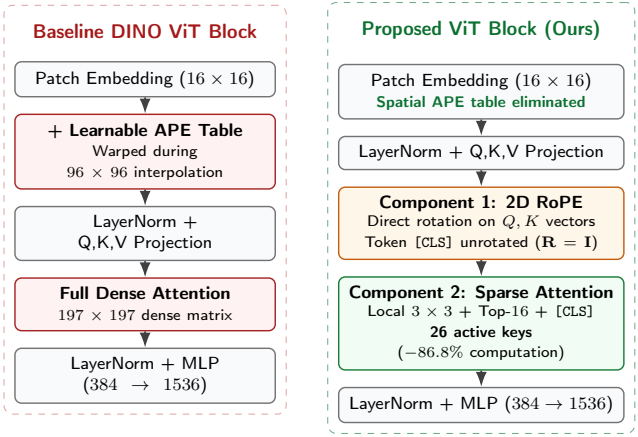
## Bottleneck 2: Full Dense Attention $\mathcal{O}(N^2)$

- **Computational Redundancy**: Every token attends to all $197$ tokens, yet $> 85\%$ of converged attention weights are near zero.
- **Background Clutter**: Dense attention diffuses mass into irrelevant background (sky, grass) rather than salient objects.

## Our Architectural Remedy

Replace static APE with **Frequency-Adaptive 2D RoPE** & dense attention with **Spatial-Semantic Sparse Attention**.

# 3. Architectural Overview: Baseline vs. Proposed ViT Block




## Core Design Principles

- **Strict Zero-Parameter Overhead**: Maintains exactly **21,665,664** parameters (100% match with baseline).
- **Theoretical FLOPs Parity**: **9.20 GFLOPs** at standard $224 \times 224$ input resolution.
- **Geometric & Saliency Synergy**: 2D RoPE provides continuous spatial awareness; Sparse Attention isolates the 26 most informative keys.

# 4. Component 1: Frequency-Adaptive 2D Rotary Position Embedding

## 2D Spatial Rotary Mechanism

Decompose head dimension $d_h = 64$ into spatial $x, y$ axes:

$$d_x = d_y = 32, \quad M = 16 \text{ frequency bands } \omega_m$$

Query and key vectors rotated by orthogonal matrix $\mathbf{R}_\Theta$:

$$\tilde{\mathbf{q}}_i = \mathbf{R}_\Theta(p_{x,i}, p_{y,i})\mathbf{q}_i, \quad \tilde{\mathbf{k}}_j = \mathbf{R}_\Theta(p_{x,j}, p_{y,j})\mathbf{k}_j$$

**Relative translation invariance via inner product**:

$$\langle \tilde{\mathbf{q}}_i, \tilde{\mathbf{k}}_j \rangle = \mathbf{q}_i^T \mathbf{R}_\Theta(p_{x,i} - p_{x,j},\, p_{y,i} - p_{y,j})\mathbf{k}_j$$

$\Rightarrow$ Directly encodes continuous 2D relative displacement!

## Frequency-Adaptive Coordinate Blending

Blends raw integer and normalized continuous coordinates:

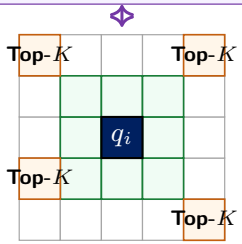
$$p_m = \alpha_m \cdot p_{\text{raw}} + (1-\alpha_m) \cdot p_{\text{norm}}, \quad \alpha_m = 1 - \frac{m}{M-1}$$

- **High Frequencies** ($\alpha \to 1$): Raw integer grid $\Rightarrow$ captures sharp local edge structures.
- **Low Frequencies** ($\alpha \to 0$): Normalized grid $\Rightarrow$ guarantees scale invariance across $224^2$ and $96^2$ crops.

**Preserving** `[CLS]` **Token**: Set $\boldsymbol{\theta}_{\text{cls}} = \mathbf{0} \Rightarrow \mathbf{R} = \mathbf{I}$, preventing artificial spatial bias on the global representation.

# 5. Component 2: Spatial-Semantic Sparse Softmax Attention

**Bidirectional `[CLS]` Conduit (Index 0)**
`[CLS]` attends to all tokens
& all tokens attend to `[CLS]`




## Active Key Set $\mathcal{S}_i$ (26 Keys Total)

- **Local $3 \times 3$ ($\mathcal{N}_{\text{local}}$)**: 9 spatial neighbors (Chebyshev radius $R = 1$).
- **Semantic Top-16 ($\mathcal{N}_{\text{semantic}}$)**: 16 highest dot-product keys across the grid.
- **Bidirectional `[CLS]`**: Seamless global-to-local communication.

**86.8% Computation Reduction** vs. dense attention ($26/197$).

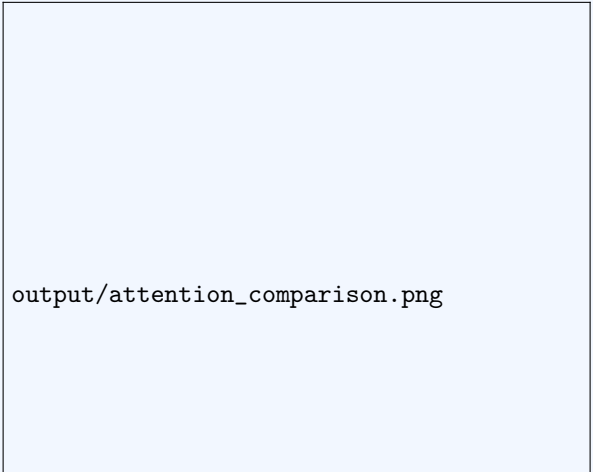
## Exact Softmax Normalization

$$\mathbf{A}_{ij} = \frac{\exp(\tilde{\mathbf{q}}_i^T \tilde{\mathbf{k}}_j / \sqrt{d_h})}{\sum_{l \in \mathcal{S}_i} \exp(\tilde{\mathbf{q}}_i^T \tilde{\mathbf{k}}_l / \sqrt{d_h})} \quad (j \in \mathcal{S}_i)$$

**Exact Formulation**: Probabilities strictly sum to 1 over $\mathcal{S}_i$, ensuring stable gradients.

# 6. Visual Evidence: Attention Maps & Empirical Gains

## 6-Head Attention Map Comparison (Epoch 190)




## Quantitative Improvements (CIFAR-10)

- **k-NN Top-1** ($k = 10$):
  $74.41\% \rightarrow \mathbf{75.97}\%$ (**+1.56% gain**).
- **Linear Probing Top-1**:
  $81.46\% \rightarrow \mathbf{83.98}\%$ (**+2.52% gain**).
- **Early Convergence Speed**:
  Epoch 10: $73.76\% \rightarrow \mathbf{79.86}\%$ (**+6.10% boost**).